Experiment Report

Aim:

To configure a GPIO pin of STM32F446RE as digital output and verify LED blinking operation using software delay routines.

Apparatus Required:

STM32 Nucleo-F446RE development board, USB TypeA to MiniB cable, STM32 CUBE IDE, STM32 CubeMX.

Theory:

Microcontrollers utilize GPIO (General-Purpose Input/Output) pins for interfacing with external devices such as LEDs, switches, or sensors. In STM32 microcontrollers, GPIO pins can be configured via software using STM32CubeIDE or STM32CubeMX as inputs or outputs.

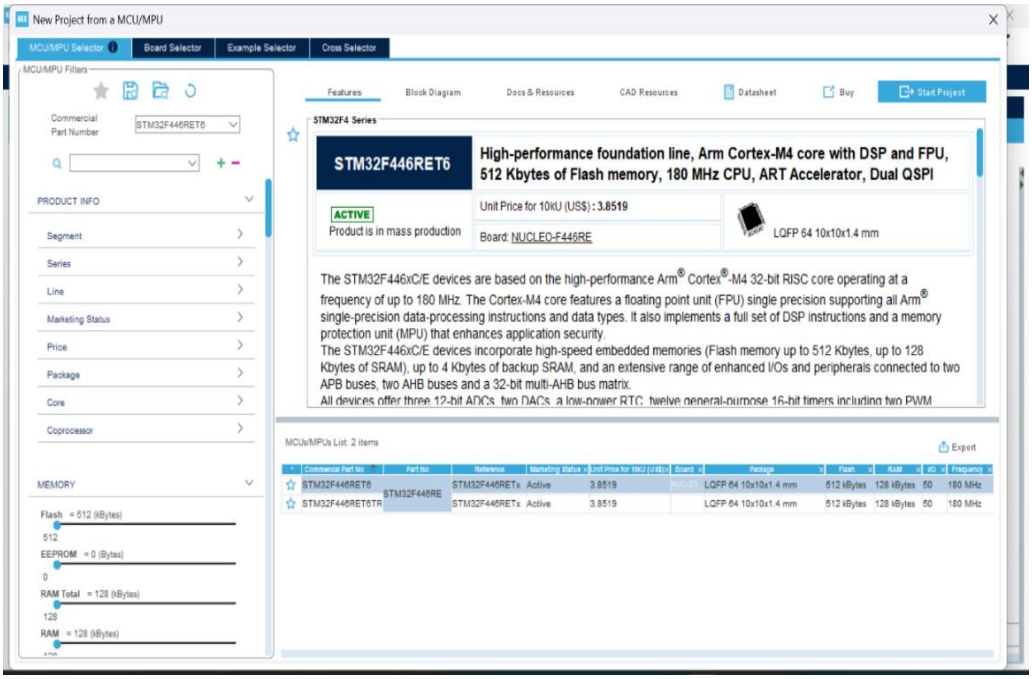
When a GPIO pin is configured for digital output, digital output of the GPIO pin can be set HIGH (logic 1) or LOW (logic 0). Therefore, if the GPIO pin is connected to an LED, then based on the GPIO pin's output voltage, the LED will light up or be off.

In this particular experiment, the method used to blink the LED is accomplished by continuously toggling the GPIO output state with a time delay specified between each toggle (ON and OFF). The delay timer is accomplished with software routines to create an interval between ON and OFF states of the LED to produce a blink that can be seen with the human eye.

Understanding how to configure GPIO pins, control digital outputs using an LED and use programmed delays for controlling timing in embedded systems through this experiment provides a basic level of understanding.

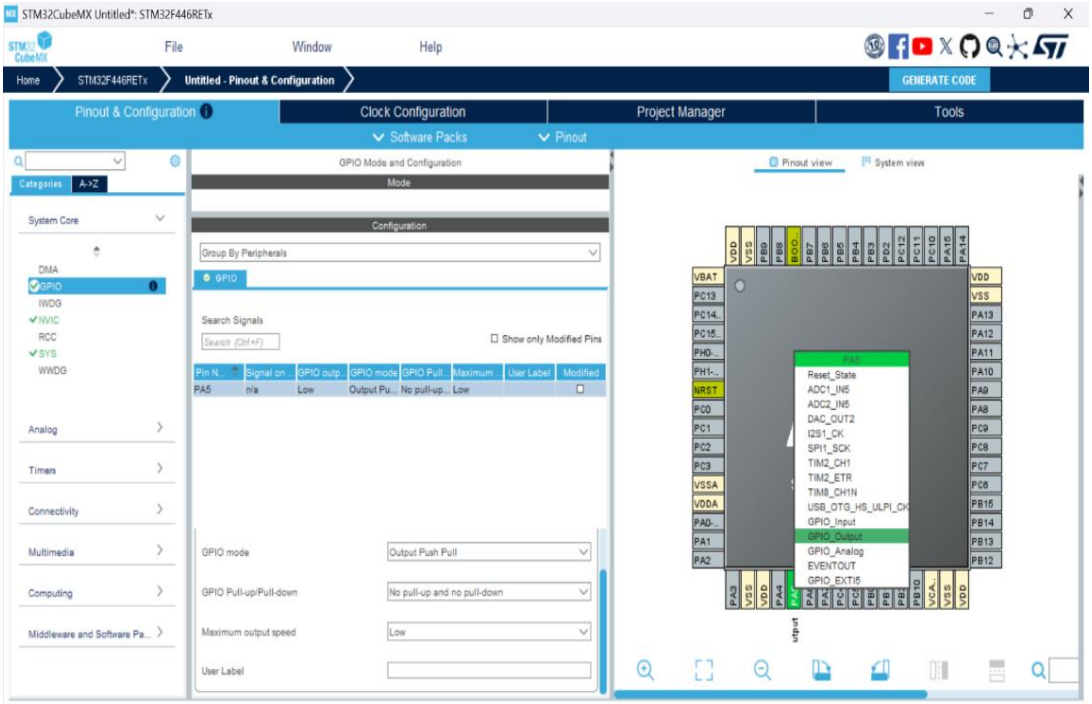
Procedure:

1. Open CubeMX and Click on ACCESS TO MCU SELECTOR.

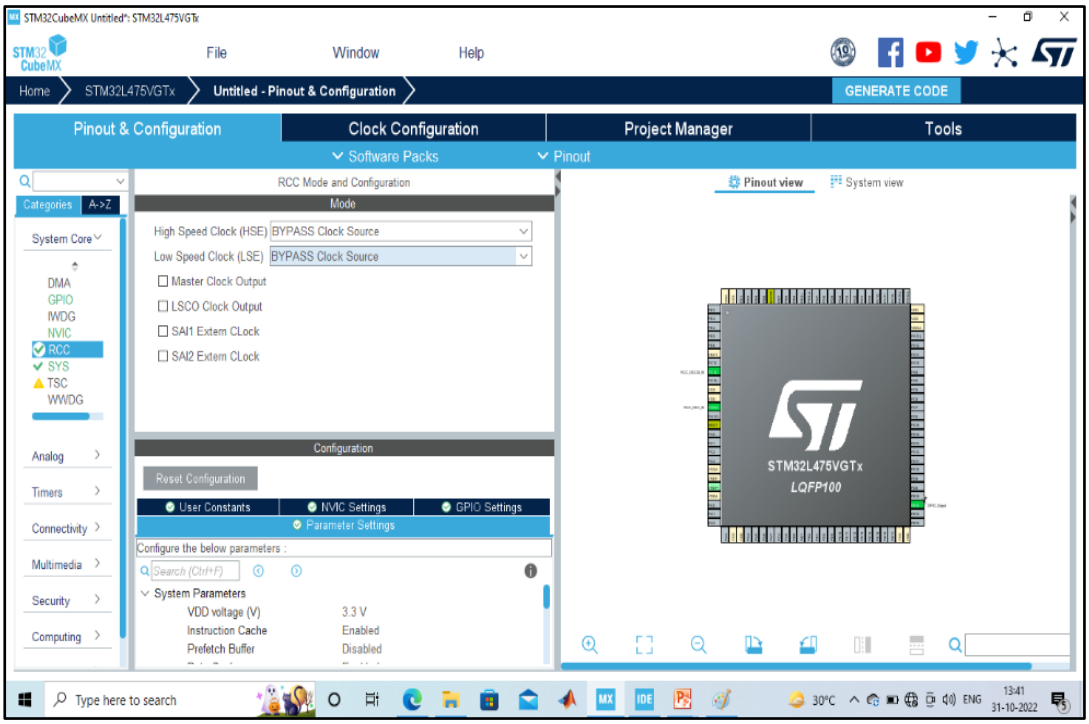


2. Write and select the microcontroller as you want to use and then click start the project. Select commercial part number STM32F446RE.

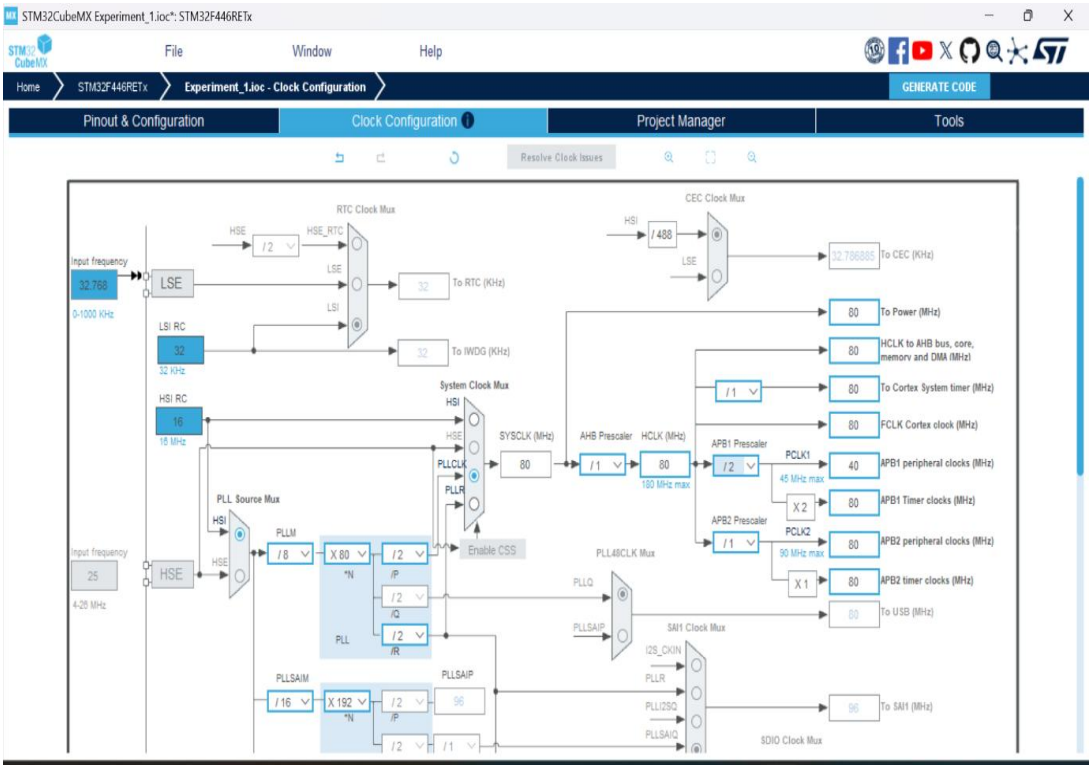
3. In system Core select GPIO and right click on pin PA5 to make it as GPIO output pin as shown below as LED (Inbuilt LED) relates to it in the board.



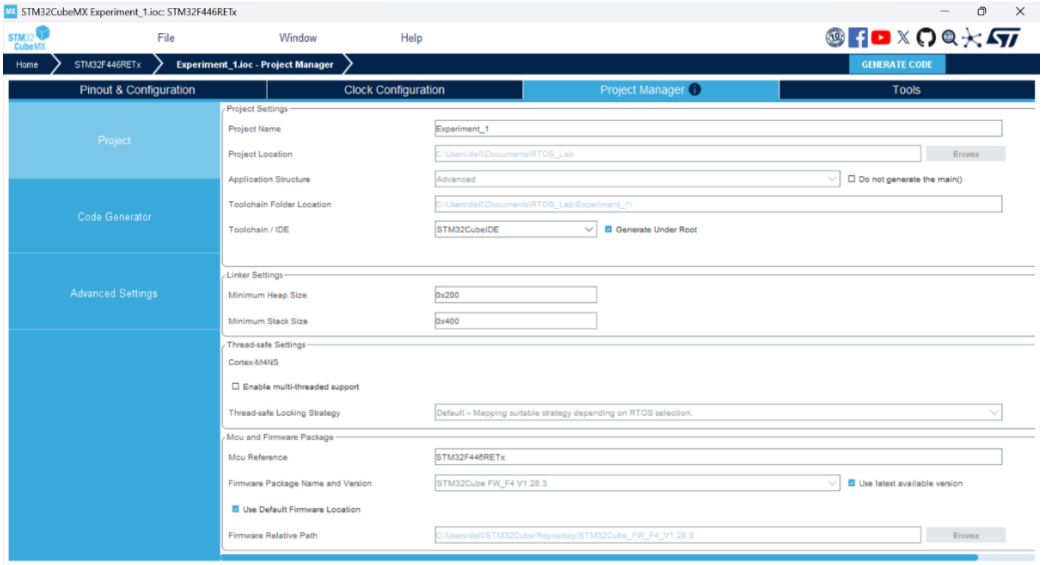
4. **Clock Selection**: Select the internal clock by clicking RCC and selecting BYPASS clock source



5. **Clock Configuration:** Configure clock as shown below to get maximum internal clock frequency.



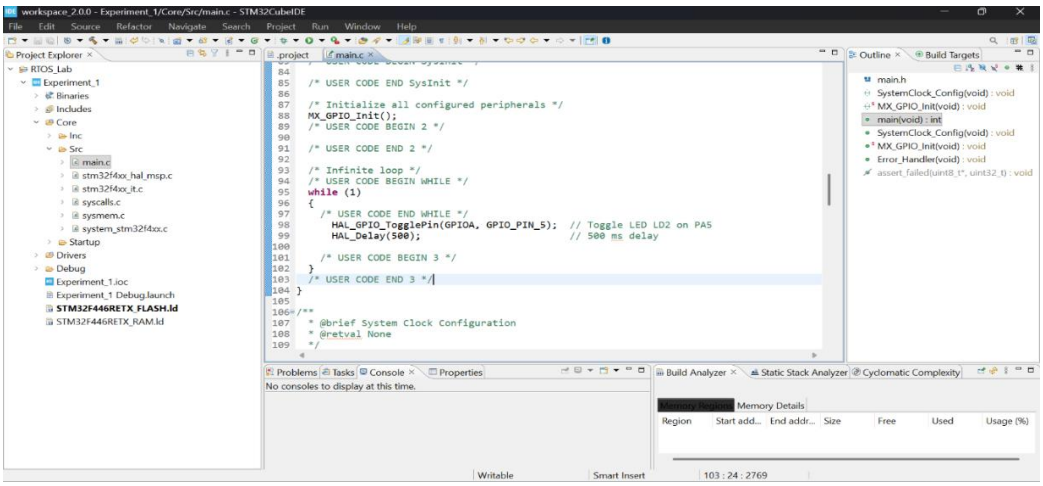
6. Write Project name and select IDE.



7. Click on generate code.
8. Click on open project.
9. Click ‘OK’ on the generated popup showing “successfully imported the project on the workspace” and the project will be successfully added to the cube IDE.
10. Open the main source code on Cube IDE by clicking on **main.c**
11. To generate bin file and hex file, right click on project and click properties.
12. Expand C/C++ Build, click on setting, click on MCU post build outputs and select convert to binary file as well as convert to hex file and click “Apply and Close ".
13. After configuration write only below two instructions in the user while loop and compile the program and ensure there will be zero error after compiling.

    HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);

    HAL_Delay(500);



14. To blink the connected LED on the board, connect the board, click on build, click ‘ok’ and click ‘switch’ on popups.
15. Now you are in debugging mode, click ‘Resume’ to execute the program.

Code:

Attached the main.c file which include full code with the experiment.

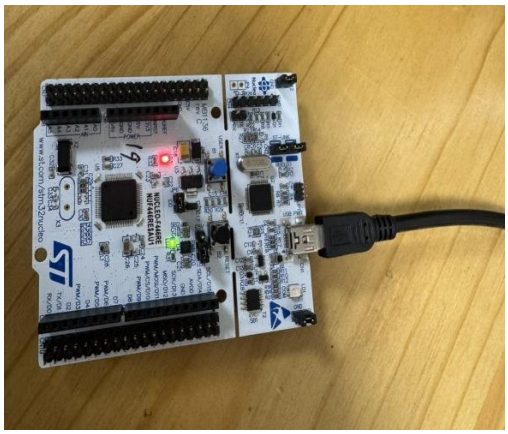
Observation:

Change the delay value (e.g., 200 ms, 1000 ms), re-build, re-run, and note the change in blink speed and fill the observation table.

S. No.	GPIO pin used	Mode / Type / Pull	Delay value (ms)	Approx. blink period (s)	Comment (slow/fast/visible)
1	PA5	Output, PP, No PU/PD	100	0.2	Fast
2	PA5	Output, PP, No PU/PD	200	0.4	Fast
3	PA5	Output, PP, No PU/PD	500	1.0	Fast
4	PA5	Output, PP, No PU/PD	1000	2.0	Slow

Output:

The onboard LED connected to PA5 blinks at a rate depending on the delay value.



Result:

The GPIO pin of STM32F446RE was successfully configured as a digital output. The LED connected to the GPIO pin blinked at a regular interval using software delay, confirming correct operation of GPIO output and timing control.